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Brake device for vehicle

Abstract

A brake device includes: a brake pedal including a pedal and a lever that rotates about a rotation axis when the pedal is operated; a housing that rotatably supports the lever; a reaction force generator that generates a reaction force to the lever in accordance with a stroke amount of the brake pedal by receiving a force from the lever when the brake pedal is operated; a deformation member including a deformation portion that is deformed by a force received from the lever when the brake pedal is operated in a state where the reaction force generated by the reaction force generator is generated; and a stroke sensor that outputs a signal corresponding to the stroke amount and outputs a value indicating that operation of the brake pedal is abnormal and instructing execution of control for stopping a vehicle when the deformation portion is deformed.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) The present application is a continuation application of U.S. application Ser. No. 17/867,860, filed Jul. 19, 2022, which is a continuation of International Patent Application No. PCT/JP2021/000520 filed on Jan. 8, 2021, which designated the U.S. and claims the benefit of priority from Japanese Patent Application No. 2020-007568 filed on Jan. 21, 2020. The entire disclosures of all of the above applications are incorporated herein by reference.

TECHNICAL FIELD

(1) The present disclosure relates to a brake device for a vehicle.

(2) A brake device for a vehicle includes: a brake pedal including a pedal and a lever that rotates about a rotation axis when the pedal is operated; and a housing that rotatably supports the lever. A detection lever detects an abnormal rotational stroke of the pedal.

SUMMARY

(3) According to an aspect of the present disclosure, a brake device for a vehicle includes: a brake pedal including a pedal and a lever that rotates about a rotation axis when the pedal is operated; a housing that rotatably supports the lever; a reaction force generator that generates a reaction force to the lever in accordance with a stroke amount of the brake pedal by receiving a force from the lever when the brake pedal is operated; a deformation member including a deformation portion that is deformed by a force received from the lever when the brake pedal is operated in a state where the reaction force generated by the reaction force generator is generated; and a stroke sensor that outputs a signal corresponding to the stroke amount and outputs a value indicating that operation of the brake pedal is abnormal and instructing execution of control for stopping the vehicle when the deformation portion is deformed.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a vehicle brake system using a vehicle brake device according to a first embodiment.
- (2) FIG. 2 is a sectional view of the vehicle brake device.
- (3) FIG. 3 is a diagram of a vehicle brake device attached to a vehicle.
- (4) FIG. 4 is a perspective view of a deformation member of the vehicle brake device.
- (5) FIG. 5 is a chart illustrating a relationship between a stroke amount, a reaction force, and a sensor output.
- (6) FIG. 6 is a sectional view of the vehicle brake device.
- (7) FIG. 7 is a sectional view of the vehicle brake device.
- (8) FIG. 8 is a sectional view of the vehicle brake device.
- (9) FIG. 9 is a sectional view of the vehicle brake device.
- (10) FIG. 10 is a sectional view of the vehicle brake device.
- (11) FIG. 11 is a sectional view of the vehicle brake device.
- (12) FIG. 12 is a flowchart illustrating processing by an ECU of the vehicle brake system.
- (13) FIG. 13 is a sectional view of a vehicle brake device according to a second embodiment.
- (14) FIG. 14 is a chart illustrating a relationship between a stroke amount, a reaction force, and a sensor output.
- (15) FIG. 15 is a sectional view of the vehicle brake device.
- (16) FIG. 16 is a sectional view of the vehicle brake device.
- (17) FIG. 17 is a sectional view of a vehicle brake device according to a third embodiment.
- (18) FIG. 18 is a sectional view of a vehicle brake device according to a fourth embodiment.
- (19) FIG. 19 is a perspective view of a deformation member of the vehicle brake device.
- (20) FIG. 20 is a sectional view of the vehicle brake device.
- (21) FIG. 21 is a sectional view of a vehicle brake device according to a fifth embodiment.
- (22) FIG. 22 is a perspective view of a deformation member of the vehicle brake device.
- (23) FIG. 23 is a chart illustrating a relationship between a stroke amount, a reaction force, and a sensor output in a vehicle brake device according to a sixth embodiment.
- (24) FIG. 24 is a chart illustrating a relationship between a stroke amount, a reaction force, and a sensor output in a vehicle brake device according to a seventh embodiment.
- (25) FIG. 25 is a chart illustrating a relationship between a stroke amount, a reaction force, and a sensor output in a vehicle brake device according to an eighth embodiment.

(26) FIG. 26 is a chart illustrating a relationship between a stroke amount, a reaction force, and a sensor output in a vehicle brake device according to a ninth embodiment.

(27) FIG. 27 is a chart illustrating a relationship between a stroke amount, a reaction force, and a sensor output in a vehicle brake device according to a tenth embodiment.

(28) FIG. 28 is a sectional view of a vehicle brake device according to another embodiment.

DESCRIPTION OF EMBODIMENTS

(29) To begin with, examples of relevant techniques will be described.

(30) Conventionally, there has been known a vehicle pedal device provided with a weak point having a supporting sectional area reduced with respect to a load applied to a pedal. In the vehicle pedal device, since the weak point is formed in a detection lever of a pedal sensor integrated with the pedal, the detection lever can be relatively easily deformed with the weak point as a starting point. As a result, the detection lever detects an abnormal rotational stroke of the pedal.

(31) According to the study of the inventors, in the vehicle pedal device, when the detection lever of the pedal sensor is deformed with the weak point as a starting point, output of the pedal sensor becomes relatively small to prevent, for example, an engine of a vehicle from being in a high output state. Here, for example, in a brake of a vehicle equipped with a brake-by-wire system, the vehicle can be stopped upon detection that a pedal is stepped on. Therefore, in the brake of the vehicle, the output of the pedal sensor needs to be relatively large such that the vehicle can stop even when the detection lever of the pedal sensor is deformed.

(32) As a result, when the vehicle pedal device is applied to a brake of a vehicle, the output of the pedal sensor becomes relatively small when the detection lever is deformed, and it may be detected that the pedal is not stepped on. Thus, the vehicle may not be able to be stopped by the brake, and safety of the vehicle may be deteriorated.

(33) The present application provides a brake device for a vehicle to improve a safety of the vehicle.

(34) According to an aspect of the present disclosure, a brake device for a vehicle includes: a brake pedal including a pedal and a lever that rotates about a rotation axis when the pedal is operated; a housing that rotatably supports the lever; a reaction force generator that generates a reaction force to the lever in accordance with a stroke amount of the brake pedal by receiving a force from the lever when the brake pedal is operated; a deformation member including a deformation portion that is deformed by a force received from the lever when the brake pedal is operated in a state where the reaction force generated by the reaction force generator is generated; and a stroke sensor that outputs a signal corresponding to the stroke amount and outputs a value indicating that operation of the brake pedal is abnormal and instructing execution of control for stopping the vehicle when the deformation portion is deformed.

(35) Thus, the vehicle can be stopped when an operation of the brake pedal is abnormal. Accordingly, the safety of the vehicle can be improved.

(36) The reference numerals attached to the components and the like indicate an example of correspondence between the components and the like and specific components and the like described in embodiments to be described below.

(37) Hereinafter, embodiments will be described with reference to the drawings. In the following embodiments, the same or equivalent portions are denoted by the same reference signs, and the description thereof will be omitted.

First Embodiment

(38) A vehicle brake device **80** is used in a vehicle brake system **1** that controls a front left wheel FL, a front right wheel FR, a rear left wheel RL, and a rear right wheel RR, which are wheels of a vehicle **6**. First, the vehicle brake system **1** will be described.

(39) As illustrated in FIG. 1, the vehicle brake system **1** includes a front left wheel cylinder, a front right wheel cylinder, a rear left wheel cylinder, and a rear right wheel cylinder. The vehicle brake system **1** includes a first actuator **10**, a second actuator **20**, a power supply **40**, an ECU **53**, and the

vehicle brake device **80**. Hereinafter, the wheel cylinder is referred to as W/C for convenience. The ECU is an abbreviation for electronic control unit.

(40) A front left W/C **2** is disposed on the front left wheel FL. A front right W/C **3** is disposed on the front right wheel FR. A rear left W/C **4** is disposed on the rear left wheel RL. A rear right W/C **5** is disposed on the rear right wheel RR. The front left W/C **2**, the front right W/C **3**, the rear left W/C **4**, and the rear right W/C **5** are connected to brake pads (not illustrated) of the vehicle **6**.

(41) The first actuator **10** includes, for example, a reservoir, a pump, a motor, a pressure sensor, and the like (not illustrated) and generates brake fluid pressure. The first actuator **10** increases the generated brake fluid pressure. Brake fluid with the increased brake fluid pressure flows to the second actuator **20** described later.

(42) The second actuator **20** includes, for example, a differential pressure control valve, a pressure increasing control valve, a pressure reducing control valve, a pump, a motor, a pressure sensor, and the like (not illustrated) and generates brake fluid pressure. The second actuator **20** causes the brake fluid flowing from the first actuator **10** to flow to each of the front left W/C **2**, the front right W/C **3**, the rear left W/C **4**, and the rear right W/C **5**.

(43) The power supply **40** supplies power to the ECU **53**.

(44) The ECU **53** mainly includes a microcomputer and the like and includes a CPU, a ROM, a RAM, a flash memory, an I/O, a bus line connecting these components, and the like. The ECU **53** controls the first actuator **10** and the second actuator **20** by executing a program stored in the ROM. Details of the control of the first actuator **10** and the second actuator **20** will be described later. The ROM, the RAM, and the flash memory are non-transitory tangible storage media.

(45) As illustrated in FIG. 2, the vehicle brake device **80** includes a brake pedal **81**, a stroke sensor **86**, a housing **88**, a reaction force generator **90**, a deformation member **95**, and a stopper **99**.

(46) The brake pedal **81** is operated by being stepped on by a driver of the vehicle **6**. Specifically, the brake pedal **81** includes a pedal **811** and a lever **812**.

(47) The pedal **811** is stepped on by the driver of the vehicle **6**.

(48) The lever **812** is connected to the pedal **811**, and rotates about a rotation axis O when the pedal **811** is stepped on by the driver of the vehicle **6**.

(49) The stroke sensor **86** is disposed, for example, on the rotation axis O of the lever **812**. As illustrated in FIG. 1, the stroke sensor **86** outputs, to the ECU **53**, a signal corresponding to a stroke amount X which is an operation amount of the brake pedal **81** by a pedal effort of the driver of the vehicle **6**. Here, the stroke amount X is, for example, a translational movement amount of the pedal **811** toward the front of the vehicle **6** as illustrated in FIG. 2. The stroke amount X and a sensor output Vs of the stroke sensor **86** are adjusted to have a linear relationship. Here, as the stroke amount X increases, the sensor output Vs of the stroke sensor **86** increases. Note that, here, the sensor output Vs is represented by, for example, a voltage. The stroke sensor **86** may output a signal corresponding to a rotation angle θ about the rotation axis O of the lever **812** to the ECU **53**. In this case, the rotation angle θ and the signal of the stroke sensor **86** are adjusted to have a linear relationship, similarly to the relationship between the stroke amount X and the sensor output Vs.

(50) As illustrated in FIGS. 2 and 3, the housing **88** is attached to a dash panel **9** which is a partition wall that separates a vehicle interior **8** from a vehicle exterior **7** of the engine room or the like of the vehicle **6**. The dash panel **9** may be referred to as a bulkhead. In the vehicle exterior **7**, not only an engine of the vehicle **6** but also a battery, an air conditioner, and the like of the vehicle **6** are disposed.

(51) As illustrated in FIG. 2, the housing **88** has a bottomed cylindrical shape, and includes a first attachment portion **881**, a second attachment portion **882**, a bolt **887**, a housing bottom **883**, a housing cylindrical portion **884**, and a housing extension **888**. Here, for convenience of description, an upper side with respect to the front of the vehicle **6** will be simply referred to as an upper side. A lower side with respect to the front of the vehicle **6** is simply referred to as a lower side.

(52) The first attachment portion **881** is connected to the housing bottom **883** described later and

extends upward from the housing bottom **883**. The first attachment portion **881** includes a first attachment hole **885**. By inserting the bolt **887** into the first attachment hole **885** and a first hole **901** of the dash panel **9**, the first attachment portion **881** is attached to the dash panel **9**. Here, the bolt **887** is inserted so as not to penetrate the dash panel **9**.

(53) The second attachment portion **882** is connected to a front side of the housing cylindrical portion **884** described later and extends downward from the housing cylindrical portion **884**. The second attachment portion **882** includes a second attachment hole **886**. By inserting the bolt **887** into the second attachment hole **886** and a second hole **902** of the dash panel **9**, the second attachment portion **882** is attached to the dash panel **9**.

(54) The housing bottom **883** supports a part of the lever **812** such that the lever **812** is rotatable about the rotation axis O, and the housing bottom **883** supports the stroke sensor **86**.

(55) The housing cylindrical portion **884** has a cylindrical shape, is connected to the housing bottom **883**, and extends downward from the housing bottom **883**. The housing cylindrical portion **884** houses a part of the lever **812**.

(56) The housing extension **888** is connected to a rear side of the housing cylindrical portion **884**. The housing extension **888** extends from the rear side of the housing cylindrical portion **884** toward a lever rear surface **814**, here, from the rear side toward the front side of the housing cylindrical portion **884**. Therefore, the housing extension **888** is in contact with the lever rear surface **814** in an initial state. Thus, the housing extension **888** supports the lever **812** such that the lever **812** does not rotate toward the rear side in the initial state. The lever rear surface **814** is a rear surface of the lever **812** and is located on a side of the lever **812** on which the pedal **811** is located.

(57) The reaction force generator **90** generates a reaction force F_r to the lever **812** in accordance with the stroke amount X. Specifically, the reaction force generator **90** includes a first elastic member **91** and a second elastic member **92**.

(58) The first elastic member **91** is, for example, a compression coil spring. The first elastic member **91** is connected to the front side of the housing cylindrical portion **884** and a lever front surface **813**. The lever front surface **813** is located on a front surface of the lever **812** and on a side of the lever **812** opposite to the pedal **811**.

(59) The second elastic member **92** is, for example, a compression coil spring. One end of the second elastic member **92** is connected to the front side of the housing cylindrical portion **884**. The other end of the second elastic member **92** is not connected to the lever **812** and faces the lever front surface **813** in the initial state. Therefore, the other end of the second elastic member **92** is not in contact with the lever front surface **813** in the initial state. The second elastic member **92** is disposed in parallel with the first elastic member **91**. Here, an elastic modulus of the second elastic member **92** is greater than an elastic modulus of the first elastic member **91**.

(60) The deformation member **95** is separated from the housing **88** and is disposed on the front side of the housing cylindrical portion **884**. The deformation member **95** may be integrated with the housing **88**.

(61) Specifically, as illustrated in FIGS. 2 and 4, the deformation member **95** includes a base **951**, an extension **952**, and a deformation portion **953**.

(62) The base **951** is connected to the front side of the housing cylindrical portion **884** by welding, depositing, adhesion, or the like, and extends in an up-down direction. Note that the base **951** and the housing cylindrical portion **884** may be connected by inserting screws or the like into holes (not illustrated) of the base **951** and the housing cylindrical portion **884**.

(63) The extension **952** is connected to the upper side of the base **951** and an upper side of a deformation portion **953** described later, and extends in a front-rear direction. The extension **952** forms a space between the lower side of the base **951** and the lower side of the deformation portion **953**. Here, in the initial state, a distance in the front-rear direction from the base **951** to the deformation portion **953** in the space between the base **951** and the deformation portion **953** increases from the rotation axis O toward the pedal **811**, here, from the upper side to the lower side.

(64) The deformation portion **953** is connected to the rear side of the extension **952** and extends in the up-down direction and in a direction from the extension **952** toward the lever front surface **813**. The deformation portion **953** includes a contact **954**, a recess **956**, and an easily deformable portion **957**.

(65) The contact **954** is connected to the lower side of the deformation portion **953** and protrudes toward the lever front surface **813**, here, rearward. The contact **954** also includes a contact surface **955**. The contact surface **955** faces the lever front surface **813**. The contact surface **955** is not in contact with the lever front surface **813** in the initial state. A distance in the front-rear direction from the contact surface **955** to the lever front surface **813** in the initial state is longer than a distance in the front-rear direction from the other end of the second elastic member **92** to the lever front surface **813** in the initial state. The contact surface **955** is curved convexly toward the lever front surface **813** by being curved in a shape of a side surface of a semicircular column. The contact surface **955** may be curved convexly toward the lever front surface **813** by being curved in a hemispherical shape.

(66) The recess **956** is formed on the upper side of the deformation portion **953**. The recess **956** is, for example, a notch and is a V-shaped groove.

(67) The easily deformable portion **957** is adjacent to the recess **956** in a depth direction of the recess **956**. Due to the recess **956**, a cross section in the depth direction of the recess **956** in the easily deformable portion **957** is smallest in cross sections in the depth direction of the recess **956** in the deformation portion **953**. Thus, when a force is applied to the deformation portion **953**, a stress applied to the easily deformable portion **957** is the greatest among stresses applied to the deformation portion **953**. Accordingly, the easily deformable portion **957** is most easily deformed in the deformation portion **953**. Therefore, the easily deformable portion **957** is likely to serve as a starting point when the deformation portion **953** is deformed.

(68) As will be described later, the stopper **99** stops the rotation of the lever **812** when the pedal **811** is abnormally operated. Specifically, as illustrated in FIG. 2, the stopper **99** is connected to the front side of the housing cylindrical portion **884**. The stopper **99** protrudes from the front side of the housing cylindrical portion **884** toward the lever front surface **813**, here, from the front side toward the rear side of the housing cylindrical portion **884**. An elastic modulus of the stopper **99** is significantly great, and is greater than the elastic modulus of the first elastic member **91**, the elastic modulus of the second elastic member **92**, and an elastic modulus of the deformation member **95**. Therefore, the stopper **99** is hardly deformed by the force received from the lever **812**. The stopper **99** includes a stopper surface **991**. The stopper surface **991** faces the lever front surface **813**. The stopper surface **991** is not in contact with the lever front surface **813** in the initial state. A distance in the front-rear direction from the stopper surface **991** to the lever front surface **813** in the initial state is longer than a distance in the front-rear direction from the contact surface **955** to the lever front surface **813** in the initial state. The stopper surface **991** is curved convexly toward the lever front surface **813** by being curved in a shape of a side surface of a semicircular column. The stopper surface **991** may be curved convexly toward the lever front surface **813** by being curved in a hemispherical shape.

(69) The vehicle brake system **1** is configured as described above. In the vehicle brake system **1**, the vehicle brake device **80** improves safety of the vehicle **6**. In order to describe this improvement in safety, operation of the vehicle brake device **80** will be described with reference to a chart illustrating a relationship between the stroke amount X , the reaction force F_r , and the sensor output V_s in FIG. 5 and to FIGS. 2 and 6 to 11.

(70) As shown in FIG. 2, before the brake pedal **81** is operated and in the initial state where the stroke amount X is 0, the lever **812** is biased by the first elastic member **91**. As a result, the housing extension **888** and the lever rear surface **814** come into contact with each other. At this time, the housing extension **888** supports the lever **812**. Therefore, the lever **812** does not rotate in the initial state. At this time, as illustrated in FIG. 5, the sensor output V_s of the stroke sensor **86** is V_{s0} which

is a value in the initial state.

(71) The brake pedal **81** is operated by the pedal effort of the driver of the vehicle **6**, and the stroke amount X becomes greater than 0 and less than $X1$. At this time, since the stroke amount X and the sensor output Vs of the stroke sensor **86** have a linear relationship, the sensor output Vs increases as the stroke amount X increases.

(72) At this time, as illustrated in FIG. **6**, a force corresponding to the pedal effort of the driver of the vehicle **6** is transmitted from the lever **812** to the first elastic member **91**. As a result, the lever **812** is separated from the housing extension **888**. The first elastic member **91** receives a force from the lever **812**. Therefore, the first elastic member **91** is compressed, and a restoring force of the first elastic member **91** is generated.

(73) At this time, the reaction force Fr to the lever **812** is generated by the restoring force of the first elastic member **91**.

(74) The restoring force of the first elastic member **91** is proportional to a deformation amount of the first elastic member **91**. The deformation amount of the first elastic member **91** is proportional to the stroke amount X . Accordingly, the restoring force of the first elastic member **91** is proportional to the stroke amount X . Thus, the stroke amount X and the reaction force Fr by the first elastic member **91** have a linear relationship.

(75) Therefore, when the stroke amount X is greater than 0 and less than $X1$, as illustrated in FIG. **5**, the reaction force Fr increases as the stroke amount X increases.

(76) When the stroke amount X is $X1$, the sensor output Vs is $Vs1$. At this time, the lever front surface **813** and the other end of the second elastic member **92** come into contact with each other. Note that $X1$ is set by a distance in the front-rear direction from the other end of the second elastic member **92** to the lever front surface **813** in the initial state, and the like.

(77) The brake pedal **81** is operated by the pedal effort of the driver of the vehicle **6**, and the stroke amount X becomes greater than or equal to $X1$ and less than $X2$. At this time, since the stroke amount X and the sensor output Vs of the stroke sensor **86** have a linear relationship, the sensor output Vs increases as the stroke amount X increases.

(78) At this time, as illustrated in FIG. **7**, the force corresponding to the pedal effort of the driver of the vehicle **6** is transmitted from the lever **812** to the first elastic member **91** and the second elastic member **92**. Accordingly, since the first elastic member **91** and the second elastic member **92** are compressed, a restoring force of the first elastic member **91** and the second elastic member **92** is generated.

(79) At this time, the reaction force Fr to the lever **812** is generated by the restoring force of the first elastic member **91** and the restoring force of the second elastic member **92**.

(80) Thus, as described above, the stroke amount X and the reaction force Fr by the first elastic member **91** have a linear relationship. The restoring force of the second elastic member **92** is proportional to a deformation amount of the second elastic member **92**, similarly to the first elastic member **91**. The deformation amount of the second elastic member **92** is proportional to the stroke amount X . Accordingly, the restoring force of the second elastic member **92** is proportional to the stroke amount X . Thus, the stroke amount X and the reaction force Fr by the second elastic member **92** have a linear relationship. Therefore, the reaction force Fr to the lever **812** at this time is a superposition of the reaction force Fr by the first elastic member **91** and the reaction force Fr by the second elastic member **92**, and thus has a linear relationship.

(81) When the stroke amount X is greater than or equal to $X1$ and less than $X2$, as illustrated in FIG. **4**, the reaction force Fr to the lever **812** increases as the stroke amount X increases. Here, the elastic modulus of the second elastic member **92** is greater than the elastic modulus of the first elastic member **91**. Thus, an amount of change in the reaction force Fr with respect to the stroke amount X when the stroke amount X is greater than or equal to $X1$ and less than $X2$ is relatively great. As a result, the force corresponding to a braking force of the vehicle **6** easily acts on the driver of the vehicle **6**, and it is easy to drive the vehicle **6**.

(82) When the stroke amount X is X_2 , the sensor output V_s is V_{s2} . At this time, the lever **812** and the deformation portion **953** come into contact with each other. Specifically, the lever front surface **813** and the contact surface **955** of the deformation portion **953** come into contact with each other. Here, since the contact surface **955** has a shape of a side surface of a semicircular column and is curved convexly toward the lever front surface **813**, the lever **812** and the deformation portion **953** come into line contact with each other. Here, in the initial state, the distance in the front-rear direction from the base **951** to the deformation portion **953** in the space between the base **951** and the deformation portion **953** increases from the upper side to the lower side. Thus, when the lever **812** and the deformation portion **953** come into contact with each other, the lever **812** of the brake pedal **81** and the extension **952** do not interfere with each other. Note that X_2 is set by a distance in the front-rear direction from the contact surface **955** of the deformation portion **953** to the lever front surface **813** in the initial state, and the like. Here, the distance in the front-rear direction from the contact surface **955** to the lever front surface **813** in the initial state is longer than the distance in the front-rear direction from the other end of the second elastic member **92** to the lever front surface **813**, and thus X_2 is greater than X_1 . Here, an operation range of the brake pedal **81** when the stroke amount X is less than or equal to X_2 is a normal use range of the brake pedal **81**. As described below, the operation range of the brake pedal **81** when the stroke amount X is greater than X_2 is a range for overload detection.

(83) The brake pedal **81** is operated by the pedal effort of the driver of the vehicle **6**, and the stroke amount X becomes greater than X_2 and less than X_3 . At this time, since the stroke amount X and the sensor output V_s of the stroke sensor **86** have a linear relationship, the sensor output V_s increases as the stroke amount X increases.

(84) At this time, as illustrated in FIG. **8**, similarly to the above, the force corresponding to the pedal effort of the driver of the vehicle **6** is transmitted from the lever **812** to the first elastic member **91** and the second elastic member **92**. Thus, the reaction force F_r to the lever **812** is generated by the restoring force of the first elastic member **91** and the second elastic member **92**.

(85) At this time, the force corresponding to the pedal effort of the driver of the vehicle **6** is transmitted from the lever **812** to the deformation portion **953**. As a result, the deformation portion **953** is elastically deformed. Here, since the deformation of the deformation portion **953** is elastic deformation, when the driver of the vehicle **6** stops stepping on the pedal **811**, a shape of the deformation portion **953** returns to a shape in the initial state.

(86) Here, as described above, the easily deformable portion **957** of the deformation portion **953** is most easily deformed in the deformation portion **953**. Therefore, when receiving the force from the lever **812**, the deformation portion **953** is deflected with the easily deformable portion **957** as a starting point. Here, the extension **952** forms a space between the lower side of the base **951** and the lower side of the deformation portion **953**. Therefore, here, the deformation portion **953** deflects the space to be small with the easily deformable portion **957** as a starting point. Since the deformation portion **953** is elastically deformed, a restoring force of the deformation portion **953** is generated. The reaction force F_r to the lever **812** is generated by the restoring force of the deformation portion **953**.

(87) Therefore, at this time, the reaction force F_r to the lever **812** is generated by the restoring force of the first elastic member **91**, the restoring force of the second elastic member **92**, and the restoring force of the deformation portion **953**.

(88) The restoring force of the deformation portion **953** is proportional to a deflection amount of the deformation portion **953**. The deflection amount of the deformation portion **953** is proportional to the stroke amount X . The restoring force due to the deflection of the deformation portion **953** is proportional to the stroke amount X . Thus, the reaction force F_r by the deformation portion **953** and the stroke amount X have a linear relationship. Note that an amount of change in the reaction force F_r with respect to the stroke amount X by the deformation portion **953** at this time is set by flexural rigidity of the deformation portion **953**. Here, the flexural rigidity is a multiplication value of a

Young's modulus of the deformation portion **953** and a second moment of area of the deformation portion **953**.

(89) Thus, as described above, the stroke amount X and the reaction force F_r by the first elastic member **91** and the second elastic member **92** have a linear relationship. Therefore, the reaction force F_r to the lever **812** at this time is a superposition of the reaction force F_r by the first elastic member **91**, the reaction force F_r by the second elastic member **92**, and the reaction force F_r by the deformation portion **953**, and thus has a linear relationship.

(90) When the stroke amount X is greater than X_2 and less than X_3 , as illustrated in FIG. 5, the reaction force F_r to the lever **812** increases as the stroke amount X increases.

(91) When the stroke amount X is X_3 , the sensor output V_s is V_{s3} . At this time, the force transmitted from the lever **812** to the deformation portion **953** causes the deformation portion **953** to yield, that is, starts to be plastically deformed. Since X_3 is the stroke amount X when the deformation portion **953** yields, X_3 is set by a material, shape, and the like of the deformation portion **953**. Here, the deformation member **95** includes, for example, a ductile material.

(92) The brake pedal **81** is operated by the pedal effort of the driver of the vehicle **6**, and the stroke amount X becomes greater than or equal to X_3 and less than X_4 . At this time, since the stroke amount X and the sensor output V_s of the stroke sensor **86** have a linear relationship, the sensor output V_s increases as the stroke amount X increases.

(93) At this time, as illustrated in FIG. 9, similarly to the above, the force corresponding to the pedal effort of the driver of the vehicle **6** is transmitted from the lever **812** to the first elastic member **91** and the second elastic member **92**. As a result, the reaction force F_r to the lever **812** by the first elastic member **91** and the second elastic member **92** is generated.

(94) At this time, the force corresponding to the pedal effort of the driver of the vehicle **6** is transmitted from the lever **812** to the deformation portion **953**. As a result, the deformation portion **953** is plastically deformed. Specifically, similarly to the above, the deformation portion **953** deflects with the easily deformable portion **957** a starting point. Here, since the deformation portion **953** is plastically deformed, when the driver of the vehicle **6** stops stepping on the pedal **811**, the shape of the deformation portion **953** does not return to the shape in the initial state.

(95) Here, since the deformation of the deformation portion **953** is deflection accompanied by plastic deformation, the deformation portion **953** generates the reaction force F_r to the lever **812** by a force of reaction to the force received from the lever **812**.

(96) Therefore, at this time, the reaction force F_r to the lever **812** is generated by the restoring force of the first elastic member **91**, the restoring force of the second elastic member **92**, and the force of reaction of the deformation portion **953** to the force received by the deformation portion **953** from the lever **812**.

(97) The force of reaction of the deformation portion **953** to the force received by the deformation portion **953** from the lever **812** is proportional to the deflection amount of the deformation portion **953**. Thus, similarly to the above, the reaction force F_r by the deformation portion **953** and the stroke amount X have a linear relationship.

(98) Thus, as described above, the stroke amount X and the reaction force F_r by the first elastic member **91** and the second elastic member **92** have a linear relationship. Therefore, the reaction force F_r to the lever **812** at this time is a superposition of the reaction force F_r by the first elastic member **91**, the reaction force F_r by the second elastic member **92**, and the reaction force F_r by the deformation portion **953**, and thus has a linear relationship.

(99) When the stroke amount X is greater than or equal to X_3 and less than X_4 , as illustrated in FIG. 5, the reaction force F_r to the lever **812** increases as the stroke amount X increases.

(100) When the stroke amount X is X_4 , the sensor output V_s is a first sensor predetermined value V_{s_th1} described later. At this time, the force transmitted from the lever **812** to the deformation portion **953** damages the deformation portion **953**. Specifically, since the easily deformable portion **957** is most easily deformed in the deformation portion **953**, the easily deformable portion **957**

breaks. Thus, since **X4** is the stroke amount **X** when the deformation portion **953** is damaged, **X4** is set by a material, shape, and the like of the deformation portion **953**.

(101) The brake pedal **81** is operated by the pedal effort of the driver of the vehicle **6**, and the stroke amount **X** becomes greater than or equal to **X4** and less than **X5**. At this time, since the stroke amount **X** and the sensor output **Vs** of the stroke sensor **86** have a linear relationship, the sensor output **Vs** increases as the stroke amount **X** increases. Therefore, the sensor output **Vs** at this time is greater than the first sensor predetermined value **Vs_th1**.

(102) At this time, as illustrated in FIG. **10**, similarly to the above, the force corresponding to the pedal effort of the driver of the vehicle **6** is transmitted from the lever **812** to the first elastic member **91** and the second elastic member **92**. Thus, the reaction force **Fr** to the lever **812** by the first elastic member **91** and the second elastic member **92** is generated.

(103) At this time, since the deformation portion **953** is damaged, here, the easily deformable portion **957** is broken, the reaction force **Fr** by the deformation portion **953** is not generated.

(104) At this time, the reaction force **Fr** to the lever **812** is generated by the restoring force of the first elastic member **91** and the second elastic member **92**.

(105) Thus, as described above, the stroke amount **X** and the reaction force **Fr** by the first elastic member **91** and the second elastic member **92** have a linear relationship. Thus, the reaction force **Fr** to the lever **812** at this time is a superposition of the reaction force **Fr** by the first elastic member **91** and the reaction force **Fr** by the second elastic member **92**, and thus has a linear relationship.

(106) When the stroke amount **X** is greater than or equal to **X4** and less than **X5**, as illustrated in FIG. **5**, the reaction force **Fr** to the lever **812** increases as the stroke amount **X** increases.

(107) When the stroke amount **X** is **X5**, the sensor output **Vs** is a second sensor predetermined value **Vs_th2**. At this time, as illustrated in FIG. **11**, the lever **812** and the stopper **99** come into contact with each other. Specifically, the lever front surface **813** and the stopper surface **991** come into contact with each other. As a result, the forward rotation of the lever **812** is stopped. Therefore, here, the second sensor predetermined value **Vs_th2** is a limit value of the sensor output **Vs** when the brake pedal **81** is operated. Note that **X5** is set by a distance in the front-rear direction from the stopper surface **991** to the lever front surface **813** in the initial state, and the like. A distance in the front-rear direction from the stopper surface **991** to the lever front surface **813** in the initial state is longer than a sum of a distance in the front-rear direction from the contact surface **955** to the lever front surface **813** and the deflection amount of the deformation portion **953** in the initial state until the deformation portion **953** is damaged. Therefore, **X5** is greater than **X4**.

(108) As described above, the vehicle brake device **80** operates.

(109) Next, in order to describe the improvement of the safety of the vehicle **6** by the vehicle brake device **80**, processing of the ECU **53** of the vehicle brake system **1** based on the sensor output **Vs** from the stroke sensor **86** will be described with reference to a flowchart in FIG. **12**. Here, for example, when an ignition of the vehicle **6** is turned on, the ECU **53** executes a program stored in the ROM of the ECU **53**.

(110) In step **S100**, the ECU **53** acquires various types of information. Specifically, the ECU **53** acquires the sensor output **Vs** corresponding to the stroke amount **X** of the brake pedal **81** from the stroke sensor **86**. The ECU **53** acquires a yaw rate, acceleration, steering angle, each wheel speed, and vehicle speed of the vehicle **6** from a sensor (not illustrated) or the like.

(111) Subsequently, in step **S110**, the ECU **53** determines whether the operation of the brake pedal **81** is abnormal. Specifically, the ECU **53** determines whether the sensor output **Vs** acquired in step **S100** is greater than or equal to the first sensor predetermined value **Vs_th1**.

(112) When the sensor output **Vs** is greater than or equal to the first sensor predetermined value **Vs_th1**, the stroke amount **X** is greater than or equal to **X4** as described above, and thus the deformation portion **953** is damaged. Thus, at this time, the ECU **53** determines that the operation of the brake pedal **81** is abnormal. Thereafter, the processing proceeds to step **S130**. Therefore, here, the first sensor predetermined value **Vs_th1** is a value indicating that the operation of the

brake pedal **81** is abnormal. The first sensor predetermined value Vs_th1 is also a value for instructing execution of control for stopping the vehicle **6** in order to ensure safety of the vehicle **6**, as described later.

(113) Here, when the stroke amount X is $X5$ greater than $X4$, the lever **812** and the stopper **99** come into contact with each other to stop the forward rotation of the lever **812**. Thus, the stroke amount X does not exceed $X5$, and the sensor output Vs does not exceed the second sensor predetermined value Vs_th2 . However, when the sensor output Vs exceeds the second sensor predetermined value Vs_th2 due to a failure or the like of the brake pedal **81** and the stroke sensor **86**, the ECU **53** also determines that the operation of the brake pedal **81** is abnormal. Thereafter, the processing proceeds to step **S130**.

(114) When the sensor output Vs is less than the first sensor predetermined value Vs_th1 , the deformation portion **953** is not damaged, and thus the ECU **53** determines that the operation of the brake pedal **81** is normal. Thereafter, the processing proceeds to step **S120**.

(115) In step **S120** subsequent to step **S110**, since the operation of the brake pedal **81** is normal, the ECU **53** performs normal control on the brake of the vehicle **6**. For example, the ECU **53** controls the first actuator **10** on the basis of the sensor output Vs acquired in step **S100**. For example, the ECU **53** increases the fluid pressure of the brake fluid flowing from the first actuator **10** to the second actuator **20** as the sensor output Vs acquired in step **S100** increases. As a result, the brake fluid having a relatively great fluid pressure flows from the first actuator **10** to the second actuator **20**. At this time, the ECU **53** controls the second actuator **20**. The second actuator **20** causes the brake fluid flowing from the first actuator **10** to the second actuator **20** to flow to the front left W/C **2**, the front right W/C **3**, the rear left W/C **4**, and the rear right W/C **5**. Thus, each brake pad (not illustrated) is in frictional contact with a brake disc corresponding to each brake pad. Therefore, since the wheel corresponding to each brake disc is decelerated, the vehicle **6** decelerates. As a result, the vehicle **6** stops.

(116) The ECU **53** also performs ABS control, VSC control, and the like. ABS is an abbreviation for antilock brake system. VSC is an abbreviation for vehicle stability control.

(117) For example, the ECU **53** calculates slip ratios of the front left wheel FL, the front right wheel FR, the rear left wheel RL, and the rear right wheel RR on the basis of the wheel speeds and the vehicle speed acquired in step **S100**. Then, the ECU **53** determines whether to execute the ABS control on the basis of the slip rates. When executing the ABS control, the ECU **53** performs any of a pressure reducing mode, a holding mode, and a pressure increasing mode in accordance with the slip rates. In the pressure reducing mode, the pressure increasing control valve (not illustrated) of the second actuator **20** corresponding to a wheel to be controlled is brought into a shut-off state, and the pressure reducing control valve (not illustrated) of the second actuator **20** is brought into an appropriate communication state, and thus the W/C pressure corresponding to the wheel to be controlled is reduced. In the holding mode, the pressure increasing control valve and the pressure reducing control valve (not illustrated) of the second actuator **20** corresponding to the wheel to be controlled are brought into a shut-off state, and thus the W/C pressure corresponding to the wheel to be controlled is held. In the pressure increasing mode, the pressure reducing control valve (not illustrated) of the second actuator **20** corresponding to a wheel to be controlled is brought into a shut-off state, and the pressure increasing control valve (not illustrated) of the second actuator **20** is brought into an appropriate communication state, and thus the W/C pressure corresponding to the wheel to be controlled is increased. Since the slip rate of each wheel of the vehicle **6** is controlled in this manner, the front left wheel FL, the front right wheel FR, the rear left wheel RL, and the rear right wheel RR are prevented from being locked.

(118) For example, the ECU **53** calculates a sideslip state of the vehicle **6** on the basis of the yaw rate, the steering angle, the acceleration, the wheel speeds, the vehicle speed, and the like acquired in step **S100**. Then, the ECU **53** determines whether to execute the VSC control on the basis of the sideslip state of the vehicle **6**. When executing the VSC control, the ECU **53** selects a wheel to be

controlled for stabilizing turning of the vehicle **6** on the basis of the sideslip state of the vehicle **6**. The ECU **53** controls the second actuator **20** such that the W/C pressure corresponding to the selected wheel to be controlled increases. As a result, the brake fluid pressure of the W/C corresponding to the wheel to be controlled increases, and thus a sideslip of the vehicle **6** is suppressed. Therefore, traveling of the vehicle **6** is stabilized.

(119) In this manner, the ECU **53** performs normal control of a brake of the vehicle **6**. Thereafter, the processing returns to step **S100**. At this time, the ECU **53** may perform collision avoidance control, regenerative cooperative control, and the like on the basis of a signal from another ECU (not illustrated).

(120) In step **S130** subsequent to step **S110**, since the sensor output V_s is greater than or equal to the first sensor predetermined value V_{s_th1} and the operation of the brake pedal **81** is abnormal, the ECU **53** forcibly stops the vehicle **6** to ensure the safety of the vehicle **6**. For example, the ECU **53** controls the first actuator **10** such that the fluid pressure of the brake fluid flowing from the first actuator **10** to the second actuator **20** becomes a predetermined fluid pressure. The ECU **53** controls the second actuator **20** such that the braking force of the vehicle **6** becomes relatively great on the basis of the slip ratio of the vehicle **6** and the like. As a result, the second actuator **20** adjusts the fluid pressure of the brake fluid flowing from the first actuator **10** on the basis of the signal from the ECU **53**. The brake fluid adjusted by the second actuator **20** flows to the front left W/C **2**, the front right W/C **3**, the rear left W/C **4**, and the rear right W/C **5**. Thus, each brake pad (not illustrated) is in frictional contact with a brake disc corresponding to each brake pad. Therefore, since the wheel corresponding to each brake disc is decelerated, the vehicle **6** decelerates. As a result, the vehicle **6** is forcibly stopped.

(121) In this manner, the processing of the ECU **53** of the vehicle brake system **1** is performed. In the vehicle brake system **1**, the vehicle brake device **80** improves the safety of the vehicle **6**. Hereinafter, the improvement of the safety by the vehicle brake device **80** will be described.

(122) In the vehicle brake device **80**, when the deformation portion **953** is damaged as illustrated in FIG. **10**, the stroke amount X is greater than or equal to X_4 and less than X_5 as illustrated in FIG. **5**. At this time, the stroke sensor **86** outputs a value indicating that the operation of the brake pedal **81** is abnormal and instructing execution of control for stopping the vehicle **6**. Specifically, the stroke sensor **86** outputs the sensor output V_s that is greater than or equal to the first sensor predetermined value V_{s_th1} to the ECU **53**. As a result, the vehicle **6** can be stopped when the operation of the brake pedal **81** is abnormal. Specifically, the processing of the ECU **53** shifts from step **S110** to step **S130**, and thus the ECU **53** forcibly stops the vehicle **6** in order to ensure the safety of the vehicle **6**. Therefore, the safety of the vehicle **6** is secured, and the safety of the vehicle **6** is improved.

(123) The vehicle brake device **80** also has effects as described below. [1] The deformation portion **953** of the deformation member **95** is deformed by receiving a force from the lever **812** when the brake pedal **81** is operated in a state where the reaction force F_r by the first elastic member **91** and the second elastic member **92** is generated. As a result, the deformation portion **953** generates the reaction force F_r . At this time, the driver operating the brake pedal **81** can feel a relatively great reaction force F_r . Therefore, deformation of the first elastic member **91** and the second elastic member **92** can be restricted such that the first elastic member **91** and the second elastic member **92** are not excessively deformed when the brake pedal **81** is operated. [2] In the vehicle brake device **80**, as illustrated in FIGS. **5** and **11**, when the stroke amount X is X_5 greater than X_4 , which is the stroke amount X when the deformation portion **953** is damaged, the lever **812** and the stopper **99** come into contact with each other. As a result, the forward rotation of the lever **812** is stopped, and this stop suppresses uneasiness of the driver of the vehicle **6** caused by excessive rotation of the lever **812**. [3] The deformation member **95** is disposed on/in the housing **88**. Specifically, the deformation member **95** is disposed on the front side of the housing cylindrical portion **884**. Here, the housing **88**, which is attached to the dash panel **9**, is fixed. As a result, the deformation member

95 is fixed by the housing **88** and does not move, and therefore, fragments of the deformation portion **953** damaged by the brake pedal **81** are relatively less likely to scatter as compared with a case where the deformation member **95** is disposed on the brake pedal **81** which is moveable. [4] As illustrated in FIG. 5, when the stroke amount X is greater than or equal to X_2 , the deformation member **95** comes into contact with the lever **812**. Specifically, the contact surface **955** of the deformation portion **953** and the lever front surface **813** come into contact with each other. Here, the contact surface **955** has a shape of a side surface of a semicircular column, and is curved convexly toward the lever front surface **813**. Thus, the contact between the contact surface **955** of the deformation portion **953** and the lever front surface **813** is a line contact. As a result, a contact area between the contact surface **955** and the lever front surface **813** is reduced as compared with a case where the contact surface **955** and the lever front surface **813** are in surface contact, and thus the accuracy of a position where the contact surface **955** and the lever front surface **813** are in contact with each other is improved. In addition, since variation in the force received by the deformation portion **953** from the lever **812** is suppressed, the accuracy of the force received by the deformation portion **953** from the lever **812** is improved. Since the accuracy of the stress applied to the deformation portion **953** is improved, accuracy of the stroke amount X when the deformation portion **953** is deformed, here, X_2 , X_3 , and X_4 described above is improved. Therefore, the accuracy of the sensor output V_s corresponding to each of X_2 , X_3 , and X_4 is improved.

(124) Here, the contact surface **955** may have a hemispherical shape, and may be curved convexly toward the lever front surface **813**. In this case, the contact between the contact surface **955** of the deformation portion **953** and the lever front surface **813** is a point contact. Accordingly, the contact area between the contact surface **955** and the lever front surface **813** becomes relatively small. Therefore, similarly to the above, the accuracy of the sensor output V_s corresponding to each of X_2 , X_3 , and X_4 is improved. [5] The deformation portion **953** includes the recess **956** and the easily deformable portion **957**. Due to the recess **956**, the cross section in the depth direction of the recess **956** in the easily deformable portion **957** is smallest among the cross sections in the depth direction of the recess **956** in the deformation portion **953**. Thus, when a force is applied to the deformation portion **953**, a stress applied to the easily deformable portion **957** is the greatest among stresses applied to the deformation portion **953**. Therefore, the easily deformable portion **957** is likely to serve as a starting point when the deformation portion **953** is deformed. The accuracy of the stroke amount X when the deformation portion **953** is deformed, here, X_2 , X_3 , and X_4 is improved. Therefore, the accuracy of the sensor output V_s corresponding to each of X_2 , X_3 , and X_4 is improved.

Second Embodiment

(125) In a second embodiment, a configuration of the deformation member is different. When the sensor output V_s is greater than or equal to the first sensor predetermined value V_{s_th1} , the stroke amount X is different. The other configurations are similar to those of the first embodiment.

(126) As illustrated in FIG. 13, the deformation member **96** is a compression coil spring. The one end of the deformation member **96** is connected to the front side of the housing cylindrical portion **884**. The other end of the deformation member **96** is not connected to the lever **812** in the initial state and faces the lever front surface **813**. Therefore, the other end of the deformation member **96** is not in contact with the lever front surface **813** in the initial state. A distance in the front-rear direction from the other end of the deformation member **96** to the lever front surface **813** in the initial state is longer than the distance in the front-rear direction from the other end of the second elastic member **92** to the lever front surface **813** in the initial state. The distance in the front-rear direction from the stopper surface **991** to the lever front surface **813** in the initial state is longer than the distance in the front-rear direction from the other end of the deformation member **96** to the lever front surface **813** in the initial state.

(127) The deformation member **96** includes a deformation portion **961** having a coil shape. As will be described later, the deformation portion **961** is elastically deformed by the force received by the

deformation member **96** from the lever **812** when the brake pedal **81** is operated. Here, an elastic modulus of the deformation member **96** is greater than the elastic modulus of the first elastic member **91** and the elastic modulus of the second elastic member **92**.

(128) Next, the operation of the vehicle brake device **80** provided with the deformation member **96** will be described with reference to a chart illustrating a relationship between the stroke amount X , the reaction force F_r , and the sensor output V_s in FIG. **14**, and FIGS. **13**, **15**, and **16**.

(129) As shown in FIG. **13**, before the brake pedal **81** is operated and in the initial state where the stroke amount X is 0, the lever **812** is biased by the first elastic member **91**, similarly to the above. As a result, the housing extension **888** and the lever rear surface **814** come into contact with each other. At this time, the housing extension **888** supports the lever **812**. Therefore, the lever **812** does not rotate in the initial state. At this time, as illustrated in FIG. **14**, the sensor output V_s of the stroke sensor **86** is V_{s0} which is a value in the initial state.

(130) The brake pedal **81** is operated by the pedal effort of the driver of the vehicle **6**, and the stroke amount X becomes greater than 0 and less than X_6 . At this time, similarly to the above, since the stroke amount X and the sensor output V_s of the stroke sensor **86** have a linear relationship, the sensor output V_s increases as the stroke amount X increases.

(131) At this time, similarly to the above, the force corresponding to the pedal effort of the driver of the vehicle **6** is transmitted from the lever **812** to the first elastic member **91**. As a result, the reaction force F_r to the lever **812** by the restoring force of the first elastic member **91** is generated.

(132) When the stroke amount X is X_6 , the lever front surface **813** and the other end of the second elastic member **92** come into contact with each other. Note that similarly to X_1 described above, X_6 is set by the distance in the front-rear direction from the other end of the second elastic member **92** to the lever front surface **813**, and the like in the initial state.

(133) The brake pedal **81** is operated by the pedal effort of the driver of the vehicle **6**, and the stroke amount X becomes greater than or equal to X_6 and less than X_7 . At this time, similarly to the above, as the stroke amount X increases, the sensor output V_s increases.

(134) At this time, similarly to the above, the force corresponding to the pedal effort of the driver of the vehicle **6** is transmitted from the lever **812** to the first elastic member **91** and the second elastic member **92**. Thus, the reaction force F_r to the lever **812** is generated by the restoring force of the first elastic member **91** and the second elastic member **92**. Here, the elastic modulus of the second elastic member **92** is greater than the elastic modulus of the first elastic member **91**. Thus, an amount of change in the reaction force F_r with respect to the stroke amount X when the stroke amount X is greater than or equal to X_7 and less than X_8 is relatively great. As a result, the force corresponding to a braking force of the vehicle **6** easily acts on the driver of the vehicle **6**, and it is easy to drive the vehicle **6**.

(135) When the stroke amount X is X_7 , the lever **812** and the deformation member **96** come into contact with each other. Specifically, the lever front surface **813** and the other end of the deformation member **96** come into contact with each other. Note that X_7 is set by a distance in the front-rear direction from the other end of the deformation member **96** to the lever front surface **813**, and the like in the initial state. Here, the operation range of the brake pedal **81** when the stroke amount X is less than or equal to X_7 is a normal use range of the brake pedal **81**. As described below, the operation range of the brake pedal **81** when the stroke amount X is greater than X_7 is a range for overload detection.

(136) The brake pedal **81** is operated by the pedal effort of the driver of the vehicle **6**, and the stroke amount X becomes greater than X_7 and less than X_8 . At this time, similarly to the above, as the stroke amount X increases, the sensor output V_s increases.

(137) At this time, similarly to the above, the force corresponding to the pedal effort of the driver of the vehicle **6** is transmitted from the lever **812** to the first elastic member **91** and the second elastic member **92**. As a result, the reaction force F_r to the lever **812** by the restoring force of the first elastic member **91** and the second elastic member **92** is generated.

(138) At this time, as illustrated in FIG. 15, a force corresponding to the pedal effort of the driver of the vehicle **6** is transmitted from the lever **812** to the deformation member **96**. As a result, the deformation portion **961** of the deformation member **96** is compressed and elastically deformed. Therefore, a restoring force of the deformation portion **961** is generated. The reaction force F_r to the lever **812** is generated by the restoring force of the deformation portion **961**.

(139) At this time, the reaction force F_r to the lever **812** is generated by the restoring force of the first elastic member **91**, the second elastic member **92**, and the deformation portion **961**.

(140) The restoring force of the deformation portion **961** is proportional to a deformation amount of the deformation portion **961**. The deformation amount of the deformation portion **961** is proportional to the stroke amount X . The restoring force of the deformation portion **961** is proportional to the stroke amount X . Thus, the reaction force F_r by the deformation portion **961** and the stroke amount X have a linear relationship.

(141) Thus, as described above, the stroke amount X and the reaction force F_r by the first elastic member **91** and the second elastic member **92** have a linear relationship. Therefore, the reaction force F_r to the lever **812** at this time is a superposition of the reaction force F_r by the first elastic member **91**, the reaction force F_r by the second elastic member **92**, and the reaction force F_r by the deformation portion **961**, and thus has a linear relationship.

(142) When the stroke amount X is greater than or equal to X_7 and less than X_8 , as illustrated in FIG. 14, the reaction force F_r to the lever **812** increases as the stroke amount X increases. Here, the elastic modulus of the deformation member **96** is greater than the elastic modulus of the first elastic member **91** and the elastic modulus of the second elastic member **92**. Thus, an amount of change in the reaction force F_r with respect to the stroke amount X when the stroke amount X is greater than or equal to X_7 and less than X_8 is sufficiently greater than an amount in change of the reaction force F_r with respect to the stroke amount X when the stroke amount X is less than X_7 . As a result, when the stroke amount X reaches X_7 or greater, the driver of the vehicle **6** can feel the brake pedal **81** hitting the deformation member **96**.

(143) When the stroke amount X is X_8 , the sensor output V_s is the first sensor predetermined value V_{s_th1} . At this time, similarly to the above, the force corresponding to the pedal effort of the driver of the vehicle **6** is transmitted from the lever **812** to the first elastic member **91**, the second elastic member **92**, and the deformation member **96**. Thus, the reaction force F_r to the lever **812** by the first elastic member **91**, the second elastic member **92**, and the deformation member **96** is generated. Therefore, the reaction force F_r to the lever **812** at this time is a superposition of the reaction force F_r by the first elastic member **91**, the reaction force F_r by the second elastic member **92**, and the reaction force F_r by the deformation portion **961**.

(144) The brake pedal **81** is operated by the pedal effort of the driver of the vehicle **6**, and the stroke amount X becomes greater than or equal to X_8 and less than X_9 . At this time, similarly to the above, as the stroke amount X increases, the sensor output V_s increases.

(145) At this time, similarly to the above, the force corresponding to the pedal effort of the driver of the vehicle **6** is transmitted from the lever **812** to the first elastic member **91**, the second elastic member **92**, and the deformation member **96**. Thus, the reaction force F_r to the lever **812** by the first elastic member **91**, the second elastic member **92**, and the deformation member **96** is generated.

(146) When the stroke amount X is X_9 , the sensor output V_s is the second sensor predetermined value V_{s_th2} . At this time, as illustrated in FIG. 16, the lever **812** and the stopper **99** come into contact with each other. Specifically, the lever front surface **813** and the stopper surface **991** come into contact with each other. As a result, the forward rotation of the lever **812** is stopped. Note that similarly to X_5 described above, X_9 is set by the distance in the front-rear direction from the stopper surface **991** to the lever front surface **813**, and the like in the initial state. The distance in the front-rear direction from the stopper surface **991** to the lever front surface **813** in the initial state is as follows. This distance is longer than a sum of the distance in the front-rear direction from the

other end of the deformation member **96** to the lever front surface **813** in the initial state and the deformation amount of the deformation portion **961** when the sensor output V_s is the first sensor predetermined value V_{s_th1} . Therefore, X_9 is greater than X_8 .

(147) As described above, the vehicle brake device **80** operates.

(148) Similarly to the above, the ECU **53** of the vehicle brake system **1** acquires the sensor output V_s corresponding to the stroke amount X from the stroke sensor **86** in step **S100**. The ECU **53** performs one of the normal control or stop control in the brake of the vehicle **6** on the basis of the acquired sensor output V_s . Specifically, when the sensor output V_s is less than the first sensor predetermined value V_{s_th1} , the operation of the brake pedal **81** is normal. Thus, the ECU **53** performs the normal control in step **S120**. When the sensor output V_s is greater than or equal to the first sensor predetermined value V_{s_th1} , the operation of the brake pedal **81** is abnormal. Thus, the ECU **53** performs the stop control in step **S130**. In this way, the safety of the vehicle **6** is ensured.

(149) In the second embodiment, effects similar to effects of the first embodiment are obtained. In the second embodiment, the deformation member **96**, which is a compression coil spring, is less likely to be damaged. Thus, effects similar to the effects described in [3] are obtained. Since the deformation member **96** is a compression coil spring, a contact area between the deformation member **96** and the lever front surface **813** is relatively small. Thus, effects similar to the effects described in [4] are obtained. Since the deformation member **96** is a compression coil spring, the deformation amount of the deformation portion **961** is in a proportional relationship with the stroke amount X , and thus is in a linear relationship with the stroke amount X . Accordingly, the accuracy of the stroke amount X when the deformation portion **961** is deformed is significantly high. Thus, effects similar to the effects described in [5] are obtained.

Third Embodiment

(150) A third embodiment is similar to the first embodiment except that the arrangement of the deformation member is different.

(151) Here, as illustrated in FIG. **17**, the deformation member **95** is disposed on the lever **812**. Here, the deformation member **95** is integrated with the lever **812**. Specifically, the base **951** of the deformation member **95** is connected to the lever front surface **813** and extends in a direction along the lever front surface **813**. Similarly to the above, the extension **952** is connected to the upper side of the base **951** and the upper side of the deformation portion **953** and extends in the front-rear direction. The deformation portion **953** of the deformation member **95** is connected to the front side of the extension **952** and extends in the up-down direction and in a direction from the extension **952** toward the front side of the housing cylindrical portion **884**. The contact **954** of the deformation portion **953** protrudes toward the front side of the housing cylindrical portion **884**, here, forward. Here, the deformation member **95** is integrated with the lever **812**, but may be separated from the lever **812**.

(152) In the third embodiment, effects similar to the effects of the first embodiment are obtained. In the third embodiment, since the deformation member **95** is disposed on the lever **812**, the effects described in [3] are not obtained.

Fourth Embodiment

(153) A fourth embodiment is similar to the first embodiment except that the configuration of the deformation member is different.

(154) Here, as illustrated in FIGS. **18** and **19**, the deformation member **95** further includes a holder **958** in addition to the base **951**, the extension **952**, the deformation portion **953**, the recess **956**, and the easily deformable portion **957** described above.

(155) The holder **958** is a leaf spring, and is connected to the lower side of the base **951** and the lower side of the deformation portion **953**. Therefore, the holder **958** is elastically deformed as the deformation portion **953** is deformed. Specifically, the holder **958** is deflected by receiving the force transmitted from the deformation portion **953**.

(156) Here, as described above, when the stroke amount X is X_4 , that is, when the sensor output V_s

is the first sensor predetermined value V_{s_th1} , the deformation portion **953** is damaged by the force transmitted from the lever **812** to the deformation portion **953**. Specifically, since the easily deformable portion **957** is most easily deformed in the deformation portion **953**, the easily deformable portion **957** breaks. At this time, as illustrated in FIG. **20**, the holder **958** receives the force transmitted from the lever **812** via the deformation portion **953**, and is deflected together with the deformation of the deformation portion **953**. Here, as described above, the holder **958** is connected to the lower side of the base **951** and the lower side of the deformation portion **953**. Therefore, even if the easily deformable portion **957** breaks and the deformation portion **953** is detached from the extension **952**, the holder **958** holds the deformation portion **953** such that the deformation portion **953** does not scatter.

(157) In the fourth embodiment, effects similar to the effects of the first embodiment are obtained.
Fifth Embodiment

(158) A fifth embodiment is similar to the fourth embodiment except that the configurations of the deformation member and the stopper are different.

(159) Here, as illustrated in FIGS. **21** and **22**, the stopper **99** is disposed on/in the deformation member **95**. Here, the stopper **99** is integrated with the deformation member **95**. In this case, the stopper **99** is connected to the base **951** of the deformation member **95** and protrudes from the base **951** towards the lever front surface **813**, here from the base **951** rearward. Similarly to the above, the stopper surface **991** of the stopper **99** faces the lever front surface **813** and is not in contact with the lever front surface **813** in the initial state. A distance in the front-rear direction from the stopper surface **991** to the lever front surface **813** in the initial state is longer than a distance in the front-rear direction from the contact surface **955** to the lever front surface **813** in the initial state. The stopper surface **991** has a shape of a side surface of a semicircular column, and is curved convexly toward the lever front surface **813**. Here, the stopper **99** is integrated with the deformation member **95**, but may be separated from the deformation member **95**.

(160) In the fifth embodiment, effects similar to the effects of the fourth embodiment are obtained.
Sixth Embodiment

(161) In a sixth embodiment, as in the first embodiment, when the stroke amount X is greater than X_2 and less than X_3 , the deformation portion **953** is elastically deformed by the force transmitted from the lever **812** to the deformation portion **953**. Here, the deformation member **95** includes, for example, a brittle material. In the sixth embodiment, as illustrated in FIG. **23**, when the stroke amount X is X_3 , the deformation portion **953** is damaged by the force transmitted from the lever **812** to the deformation portion **953**. Since the deformation portion **953** is damaged, here, the easily deformable portion **957** is broken when the stroke amount X is greater than X_3 and less than X_5 , the reaction force F_r by the deformation portion **953** is not generated. At this time, the reaction force F_r to the lever **812** is generated by the restoring force of the first elastic member **91** and the second elastic member **92**. Here, when the stroke amount X is X_4 greater than X_3 and less than X_5 , here, when the deformation portion **953** is damaged, the sensor output V_s is the first sensor predetermined value V_{s_th1} .

(162) In the sixth embodiment, effects similar to the effects of the first embodiment are obtained.
Seventh Embodiment

(163) In a seventh embodiment, as illustrated in FIG. **24**, when the stroke amount X is greater than X_2 and less than X_4 , the deformation portion **953** is elastically deformed by the force transmitted from the lever **812** to the deformation portion **953**. When the stroke amount X is X_4 , the force transmitted from the lever **812** to the deformation portion **953** causes the deformation portion **953** to yield, that is, starts to be plastically deformed. When the stroke amount X is greater than or equal to X_4 and less than X_5 , the deformation of the deformation portion **953** is deflection accompanied by plastic deformation, and thus the deformation portion **953** generates the reaction force F_r to the lever **812** by the force of reaction to the force received from the lever **812**. Therefore, at this time, the reaction force F_r to the lever **812** is generated by the restoring force of the first elastic member

91, the restoring force of the second elastic member **92**, and the force of reaction of the deformation portion **953** to the force received by the deformation portion **953** from the lever **812**. Here, when the stroke amount X is $X4$ and when the deformation portion **953** is damaged, the sensor output Vs is the first sensor predetermined value Vs_th1 .

(164) In the seventh embodiment, effects similar to the effects of the first embodiment are obtained.
Eighth Embodiment

(165) In an eighth embodiment, as in the first embodiment, the deformation portion **953** is deformed with respect to the stroke amount X , and thus the reaction force Fr and the sensor output Vs change with respect to the stroke amount X . Specifically, when the stroke amount X is $X3$, the force transmitted from the lever **812** to the deformation portion **953** causes the deformation portion **953** to yield, that is, starts to be plastically deformed. At this time, the reaction force Fr to the lever **812** is generated by the restoring force of the first elastic member **91**, the restoring force of the second elastic member **92**, and the force of reaction of the deformation portion **953** to the force received by the deformation portion **953** from the lever **812**. When the stroke amount X is $X4$, the deformation portion **953** is damaged by the force transmitted from the lever **812** to the deformation portion **953**. When the stroke amount X is $X5$, the lever **812** and the stopper **99** come into contact with each other to stop the forward rotation of the lever **812**.

(166) In the eighth embodiment, as illustrated in FIG. 25, when the stroke amount X is greater than $X4$ and less than $X5$, here, when the deformation portion **953** is damaged, the sensor output Vs is the first sensor predetermined value Vs_th1 .

(167) In the eighth embodiment, effects similar to the effects of the first embodiment are obtained. As in the eighth embodiment, the sensor output Vs when the stroke amount X is greater than $X4$ and less than $X5$ may be the first sensor predetermined value Vs_th1 .

Ninth Embodiment

(168) In a ninth embodiment, as in the first embodiment, the deformation portion **953** is deformed with respect to the stroke amount X , and thus the reaction force Fr and the sensor output Vs change with respect to the stroke amount X .

(169) In the ninth embodiment, as illustrated in FIG. 26, when the stroke amount X is greater than or equal to $X3$ and less than $X4$, here, when the deformation portion **953** is plastically deformed, the sensor output Vs is the first sensor predetermined value Vs_th1 .

(170) In the ninth embodiment, effects similar to the effects of the first embodiment are obtained. As in the ninth embodiment, the sensor output Vs when the stroke amount X is greater than or equal to $X3$ less than $X4$ and less than $X4$ may be the first sensor predetermined value Vs_th1 .

Tenth Embodiment

(171) In a tenth embodiment, as in the first embodiment, the deformation portion **953** is deformed with respect to the stroke amount X , and thus the reaction force Fr and the sensor output Vs change with respect to the stroke amount X .

(172) In the tenth embodiment, as illustrated in FIG. 27, the sensor output Vs when the stroke amount X is greater than $X2$ and less than $X3$, here, when the deformation portion **953** is elastically deformed is the first sensor predetermined value Vs_th1 .

(173) In the tenth embodiment, effects similar to effects of the first embodiment are obtained. As in the tenth embodiment, the sensor output Vs when the stroke amount X is greater than $X2$ and less than $X3$ may be the first sensor predetermined value Vs_th1 .

Other Embodiments

(174) The present disclosure is not limited to the above embodiments, and can be appropriately modified from the above embodiments. For example, the embodiments may be appropriately combined. In each of the embodiments, it goes without saying that the elements constituting the embodiments are not necessarily essential except for a case where it is explicitly stated that the elements are particularly essential and a case where the elements are considered to be obviously essential in principle.

(175) A control unit and the like and a method thereof described in the present disclosure may be achieved by a dedicated computer provided by configuring a processor and a memory programmed to execute one or a plurality of functions embodied by a computer program. Alternatively, the control unit and the like and the method thereof described in the present disclosure may be achieved by a dedicated computer provided by configuring a processor with one or more dedicated hardware logic circuits. Alternatively, the control unit and the like and the method thereof described in the present disclosure may be achieved by one or more dedicated computers configured by a combination of a processor and a memory programmed to execute one or more functions and a processor configured by one or more hardware logic circuits. The computer program may be stored in a computer-readable non-transitory tangible recording medium as an instruction executed by a computer. (1) In the embodiments, the vehicle brake device **80** includes the stroke sensor **86**. The number of stroke sensors **86** is not limited to one, and may be two or more. (2) In the embodiments, the first elastic member **91** and the second elastic member **92** are compression coil springs. However, the first elastic member **91** and the second elastic member **92** are not limited to the compression coil springs. The deformation member **95** and the stopper **99** are disposed on the front side of the housing cylindrical portion **884**. However, the first elastic member **91**, the second elastic member **92**, the deformation member **95**, and the stopper **99** are not limited to being disposed on the front side of the housing cylindrical portion **884**.

(176) For example, as illustrated in FIG. **28**, the first elastic member **91** may be a tension spring and may be connected to the rear side of the housing cylindrical portion **884** and the lever rear surface **814**. The second elastic member **92** may be a tension spring. In this case, the one end of the second elastic member **92** is connected to the rear side of the housing cylindrical portion **884**. The other end of the second elastic member **92** is connected to a second elastic member contact **921**. The second elastic member contact **921** is fixed to the housing cylindrical portion **884**. When the brake pedal **81** is operated, the second elastic member contact **921** comes into contact with the lever front surface **813**. As a result, the second elastic member contact **921** receives a force from the lever **812**. Since the second elastic member **92** is pulled by the force received by the second elastic member contact **921**, a restoring force of the second elastic member **92** is generated. The deformation member **95** has a U shape. The base **951** of the deformation member **95** is connected to the rear side of the housing cylindrical portion **884** and extends in the front-rear direction. The deformation portion **953** of the deformation member **95** is connected to the base **951** here and extends in a left-right direction with respect to the front side of the vehicle **6**. The recess **956** of the deformation portion **953** is formed opposite to the contact surface **955** of the deformation portion **953**. The recess **956** forms the easily deformable portion **957**. The stopper **99** has a U shape and is connected to the rear side of the housing cylindrical portion **884**. In such a configuration, effects similar to the effects of the first embodiment are obtained. This configuration and the first embodiment may be combined. (3) In the embodiments, the reaction force generator **90** includes one first elastic member **91** and one second elastic member **92**. However, the number of first elastic members **91** is not limited to one, and may be plural. The number of second elastic members **92** is not limited to one, and may be plural. The reaction force generator **90** may include a damper. The damper includes, for example, a viscous fluid such as oil and air. The reaction force F_r corresponding to a stroke change amount ΔX , which is an amount of change in the stroke amount X per unit time, is generated by the viscous fluid. (4) In the embodiments, the recess **956** is a V-shaped groove. However, the recess **956** is not limited to the V-shaped groove, and may be, for example, a U-shaped groove, a step, or the like. (5) In the embodiments, as the stroke amount X increases, the sensor output V_s of the stroke sensor **86** increases. However, as the stroke amount X increases, the sensor output V_s of the stroke sensor **86** is not limited to increase. For example, as the stroke amount X increases, the sensor output V_s of the stroke sensor **86** may decrease. (177) In this case, when the deformation portion **953** is deformed, for example, in the first embodiment, when the stroke amount X is greater than or equal to X_4 , the stroke sensor **86** outputs

the sensor output Vs that is less than or equal to the first sensor predetermined value Vs_th1 to the ECU 53.

(178) Here, when the stroke amount X is X5 greater than X4, the lever 812 and the stopper 99 come into contact with each other to stop the forward rotation of the lever 812. At this time, since the sensor output Vs of the stroke sensor 86 decreases as the stroke amount X increases, the sensor output Vs when the stroke amount X is X5 is less than the sensor output Vs when the stroke amount X is X4. Therefore, in this case, the second sensor predetermined value Vs_th2, which is the sensor output Vs when the stroke amount X is X5, is less than the first sensor predetermined value Vs_th1, which is the sensor output Vs when the stroke amount X is X4.

(179) In this case, when the sensor output Vs is greater than the first sensor predetermined value Vs_th1, the operation of the brake pedal 81 is normal. Thus, the ECU 53 performs the normal control in step S120. When the sensor output Vs is less than or equal to the first sensor predetermined value Vs_th1, the operation of the brake pedal 81 is abnormal. Thus, the ECU 53 performs the stop control in step S130. In this way, the safety of the vehicle 6 is ensured.

Claims

1. A brake device for a vehicle comprising: a brake pedal including a pedal and a lever that rotates about a rotation axis when the pedal is operated; a housing that rotatably supports the lever; a reaction force generator configured to generate a reaction force to the lever in accordance with a stroke amount of the brake pedal by receiving a force from the lever when the brake pedal is operated; a deformation member including a deformation portion configured to be deformed by a force received from the lever when the brake pedal is operated in a state where the reaction force generated by the reaction force generator is generated; a stroke sensor configured to output a signal corresponding to the stroke amount; and a determiner configured to determine that an operation of the brake pedal is abnormal, wherein the determiner determines that the operation of the brake pedal is abnormal when the signal output from the stroke sensor is greater than or equal to a predetermined value.
2. The brake device according to claim 1, wherein the deformation portion is configured to be damaged by the force received from the lever when the brake pedal is operated, and when the deformation portion is damaged, the determiner determines that the operation of the brake pedal is abnormal.
3. The brake device according to claim 1, wherein the deformation portion is configured to be plastically deformed by the force received from the lever when the brake pedal is operated, and when the deformation portion is plastically deformed, the determiner determines that the operation of the brake pedal is abnormal.
4. The brake device according to claim 1, wherein the deformation portion is configured to be elastically deformed by the force received from the lever when the brake pedal is operated, and when the deformation portion is elastically deformed, the determiner determines that the operation of the brake pedal is abnormal.
5. The brake device according to claim 1, further comprising a stopper configured to come into contact with the lever and stop rotation of the lever when the stroke amount is greater than the stroke amount when the deformation portion is deformed.
6. The brake device according to claim 5, wherein the stopper is disposed on the housing.
7. The brake device according to claim 5, wherein the stopper is disposed on the deformation member.
8. The brake device according to claim 1, wherein the deformation member is disposed on the housing.
9. The brake device according to claim 1, wherein the deformation portion includes a contact surface that comes into contact with the lever when the brake pedal is operated, and the contact

surface is curved convexly toward the lever.

10. The brake device according to claim 1, wherein the determiner determines that the operation of the brake pedal is abnormal when the signal corresponding to the stroke amount is greater than or equal to a predetermined value corresponding to the stroke amount when the deformation portion is deformed.

11. The brake device according to claim 1, wherein the determiner includes a processor and a memory, the processor is configured to determine that the operation of the brake pedal is abnormal when the signal output from the stroke sensor is greater than or equal to a predetermined value, and the processor is configured to output a signal indicating that operation of the brake pedal is abnormal and instruct execution of control for stopping the vehicle when it is determined that the operation of the brake pedal is abnormal.
